Training Output:

```
----------------------------------------
| time/                   |            |
|    fps                  | 507        |
|    iterations           | 90         |
|    time_elapsed         | 2906       |
|    total_timesteps      | 1474560    |
| train/                  |            |
|    approx_kl            | 0.05119119 |
|    clip_fraction        | 0.373      |
|    clip_range           | 0.2        |
|    entropy_loss         | -0.872     |
|    explained_variance   | 0.831      |
|    learning_rate        | 0.0003     |
|    loss                 | -0.0939    |
|    n_updates            | 890        |
|    policy_gradient_loss | -0.0395    |
|    value_loss           | 0.0595     |
----------------------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 506         |
|    iterations           | 91          |
|    time_elapsed         | 2940        |
|    total_timesteps      | 1490944     |
| train/                  |             |
|    approx_kl            | 0.053537175 |
|    clip_fraction        | 0.375       |
|    clip_range           | 0.2         |
|    entropy_loss         | -0.861      |
|    explained_variance   | 0.856       |
|    learning_rate        | 0.0003      |
|    loss                 | -0.0779     |
|    n_updates            | 900         |
|    policy_gradient_loss | -0.0437     |
|    value_loss           | 0.0838      |
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-----------------------------------------
| time/                   |             |
|    fps                  | 506         |
|    iterations           | 92          |
|    time_elapsed         | 2975        |
|    total_timesteps      | 1507328     |
| train/                  |             |
|    approx_kl            | 0.050909456 |
|    clip_fraction        | 0.374       |
|    clip_range           | 0.2         |
|    entropy_loss         | -0.88       |
|    explained_variance   | 0.87        |
|    learning_rate        | 0.0003      |
|    loss                 | -0.0829     |
|    n_updates            | 910         |
|    policy_gradient_loss | -0.0387     |
|    value_loss           | 0.0554      |
-----------------------------------------
 100% ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━━━━━━━━━━━ 1,507,328/1,500,000  [ 0:49:21 < 0:00:00 , 448 it/s ]
 100% ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
━━━━━━━━━━━━ 1,507,328/1,500,000  [ 0:49:21 < 0:00:00 , 547 it/s ]
[INFO    ] 13:51:35  __main__  Training complete.
[INFO    ] 13:51:35  __main__  Model saved to D:\Softwares\GitHub\quant-predator-
v2\models\eurusd\ppo_model.zip
[INFO    ] 13:51:35  __main__  VecNormalize stats saved to D:\Softwares\GitHub\quant-predator-
v2\models\eurusd\vec_normalize.pkl
```

Backtest Result:

```
[INFO    ] 13:51:35  __main__  Running out-of-sample evaluation on test set...
[INFO    ] 13:51:35  engine.rl_env  ForexTradingEnv initialised: 6,790 bars | 35 features | obs_dim=842
[INFO    ] 13:51:35  engine.backtester  OOS Test: Simulating 6,790 bars on EURUSD...
[INFO    ] 13:51:56  engine.backtester  OOS Test complete. Final balance: $7,902.14 | Trades executed:
521

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  QUANT PREDATOR v2 — OUT-OF-SAMPLE
════════════════════════════════════════════════════════════
  Starting Balance : $   10,000.00
  Ending Balance   : $    7,902.14
  Total Return     :      -20.98%
  Annual Return    :      -19.78%
────────────────────────────────────────
  Max Drawdown     :      -38.30%
  Sharpe Ratio     :      -0.388  (H1-annualised)
  Sortino Ratio    :      -0.554
  Calmar Ratio     :      -0.516
────────────────────────────────────────
  Total Trades     :         395
  Win Rate         :       26.6%
  Profit Factor    :       0.928
  Avg Win          : $   +258.47
  Avg Loss         : $   -100.82
  Expectancy       : $     -5.31
────────────────────────────────────────
  Verdict: FAILING — Not suitable for deployment.
════════════════════════════════════════════════════════════
```

```
[WARNING ] 13:51:56  __main__  VALIDATION FAILED — Sharpe: -0.39 | Max DD: 38.3%. Model saved but NOT recommended for live deployment. Consider retraining with different hyperparameters.
```